

CBSE Class 10 Mathematics — Circles — Solutions

1. 1. If the angle between two tangents drawn from a point P to a circle of radius r and center O is 60° , then the length of OP is:

Answer: (B)

Solution:

1. The line segment joining the center to the external point bisects the angle between the tangents.
2. The triangle formed by the center, the point of tangency, and the external point is a right-angled triangle.
3. Using trigonometry, $\sin(30^\circ) = r/OP$.

Derivation:

$$\begin{aligned}\sin(30^\circ) &= \frac{r}{OP} \\ \implies \frac{1}{2} &= \frac{r}{OP} \\ \implies OP &= 2r\end{aligned}$$

Circles — Tangents to a circle

2. 2. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the center O at a point Q so that $OQ = 13$ cm. The length of PQ is:

Answer: (C)

Solution:

1. The tangent at any point of a circle is perpendicular to the radius through the point of contact.
2. This forms a right-angled triangle OPQ where OQ is the hypotenuse.
3. Apply the Pythagorean theorem: $PQ^2 = OQ^2 - OP^2$.

Derivation:

$$PQ = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12 \text{ cm}$$

Circles — Tangents to a circle

3. 3. The number of tangents that can be drawn from a point lying inside a circle is:

Answer: (A)

Solution:

1. A tangent touches the circle at exactly one point.
2. If a point is inside the circle, any line passing through it will intersect the circle at two points (a secant).
3. Therefore, no tangent can be drawn from an interior point.

Derivation:

$$\text{Number of tangents} = 0$$

Circles — Tangents to a circle

4. **4.** If two tangents inclined at an angle of 80° are drawn to a circle of radius 3 cm, then the angle between the two radii at the center is:

Answer: (C)

Solution:

1. The quadrilateral formed by the center, two points of contact, and the external point has angles summing to 360° .
2. The angles at the points of contact are 90° each.
3. The angle at the center and the angle between tangents are supplementary.

Derivation:

$$\text{Angle} = 180^\circ - 80^\circ = 100^\circ$$

Circles — Tangents to a circle

5. **5.** The length of a tangent from a point A at a distance of 10 cm from the center of the circle is 8 cm. The radius of the circle is:

Answer: (B)

Solution:

1. The radius, tangent, and the line to the external point form a right triangle.
2. The distance from the center is the hypotenuse (10), and the tangent is one leg (8).
3. Use $r^2 + 8^2 = 10^2$.

Derivation:

$$\begin{aligned} r^2 &= 100 - 64 = 36 \\ \implies r &= 6 \text{ cm} \end{aligned}$$

Circles — Tangents to a circle

6. **6.** Assertion (A): The length of a tangent drawn from an external point to a circle is equal to the length of another tangent drawn from the same point. Reason (R): The lengths of tangents drawn from an external point to a circle are equal.

Answer: (A)

Solution:

1. Identify the theorem: The lengths of tangents drawn from an external point to a circle are equal.
2. Check Assertion: It states the definition of the theorem.
3. Check Reason: It states the theorem itself.
4. Conclusion: R explains A correctly.

Derivation: $A : \text{True}, R : \text{True}, R \implies A$

Circles — Tangents to a circle

7. **7.** Assertion (A): The tangent at any point of a circle is perpendicular to the radius through the point of contact. Reason (R): The perpendicular distance from the center of the circle to the tangent is equal to the radius.

Answer: (A)

Solution:

1. Recall NCERT Theorem 10.1: The tangent at any point of a circle is perpendicular to the radius through the point of contact.
2. Verify Reason: The shortest distance from the center to the tangent is the radius, identifying the perpendicularity.
3. Conclusion: Both are true and R explains A.

Derivation: Radius $\perp$ Tangent at point of contact *Circles — Tangents to a circle*

8. **8.** Assertion (A): The number of tangents that can be drawn from a point inside a circle is zero. Reason (R): A point inside a circle lies in the interior, and any line passing through it will be a secant, not a tangent.

Answer: (A)

Solution:

1. Analyze A: A point inside the circle does not permit any tangent as a tangent must touch the circle at only one point.
2. Analyze R: Any line through an interior point intersects the circle at two points (secant).
3. Conclusion: Both true and R explains A.

Derivation: No tangent from interior point. *Circles — Tangents to a circle*

9. **9.** Assertion (A): If the angle between two tangents drawn from a point P to a circle of radius r and center O is 60° , then $OP = 2r$. Reason (R): The line segment from the center to the external point bisects the angle between the tangents.

Answer: (A)

Solution:

1. In $\triangle OAP$ (where A is point of contact), $\angle OPA = 30^\circ$ (bisected angle).
2. $\sin(30^\circ) = \frac{OA}{OP} = \frac{r}{OP}$.
3. $\frac{1}{2} = \frac{r}{OP} \implies OP = 2r$.
4. Conclusion: Both are true and R explains A.

Derivation:

$$\begin{aligned}\sin(30^\circ) &= \frac{r}{OP} \\ \implies OP &= 2r\end{aligned}$$

Circles — Tangents to a circle

10. **10.** Assertion (A): The length of the tangent from a point A at a distance of 5 cm from the center of the circle is 4 cm. The radius of the circle is 3 cm. Reason (R): In a right triangle formed by radius, tangent, and hypotenuse (distance to center), $r^2 + t^2 = d^2$.

Answer: (A)

Solution:

1. Use Pythagoras Theorem: $r^2 + t^2 = d^2$.
2. $r^2 + 4^2 = 5^2$.
3. $r^2 + 16 = 25 \implies r^2 = 9 \implies r = 3$.
4. Conclusion: Both are true and R explains A.

Derivation:

$$3^2 + 4^2 = 9 + 16 = 25 = 5^2$$

Circles — Tangents to a circle

11. **11.** A tangent PQ at a point P of a circle of radius 5 cm meets a line through the center O at a point Q so that $OQ = 13$ cm. Find the length of PQ .

Answer:

$$12cm$$

Solution:

1. Use the property that the tangent at any point of a circle is perpendicular to the radius through the point of contact.
2. Apply the Pythagorean theorem in the right-angled triangle OPQ where $\angle OPQ = 90^\circ$.

Derivation:

$$\begin{aligned} OP^2 + PQ^2 &= OQ^2 \\ \implies 5^2 + PQ^2 &= 13^2 \\ \implies 25 + PQ^2 &= 169 \\ \implies PQ^2 &= 144 \\ \implies PQ &= 12cm \end{aligned}$$

Circles — Tangents to a circle

12. **12.** If tangents PA and PB from a point P to a circle with center O are inclined to each other at an angle of 80° , then find $\angle POA$.

Answer:

$$50^\circ$$

Solution:

1. In quadrilateral $OAPB$, $\angle OAP = \angle OBP = 90^\circ$. The sum of angles is 360° .
2. Find $\angle AOB = 180^\circ - 80^\circ = 100^\circ$. Since $\triangle OAP \cong \triangle OBP$, $\angle POA = \frac{1}{2}\angle AOB$.

Derivation:

$$\angle AOB = 180^\circ - 80^\circ = 100^\circ. \text{ Thus, } \angle POA = \frac{100^\circ}{2} = 50^\circ$$

Circles — Tangents from an external point

- 13. 13.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8cm$$

Solution:

1. Let O be the center, and AB be the chord of the larger circle tangent to the smaller circle at M . $OM \perp AB$.
2. In $\triangle OMA$, $OA = 5$ and $OM = 3$. Use Pythagoras to find AM , then $AB = 2AM$.

Derivation:

$$AM = \sqrt{OA^2 - OM^2} = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = 4. AB = 2 \times 4 = 8cm.$$

Circles — Tangents to a circle

- 14. 14.** A quadrilateral $ABCD$ is drawn to circumscribe a circle. If $AB = 6$ cm, $BC = 7$ cm, and $CD = 4$ cm, find the length of AD .

Answer:

$$3cm$$

Solution:

1. Use the property that sums of opposite sides of a quadrilateral circumscribing a circle are equal.
2. Apply $AB + CD = BC + AD$.

Derivation:

$$\begin{aligned} 6 + 4 &= 7 + AD \\ \implies 10 &= 7 + AD \\ \implies AD &= 3cm. \end{aligned}$$

Circles — Tangents to a circle

- 15. 15.** If a point P is 26 cm away from the center O of a circle and the length of the tangent PT drawn from P to the circle is 10 cm, find the radius of the circle.

Answer:

$$24cm$$

Solution:

1. In $\triangle OTP$, $\angle OTP = 90^\circ$. OT is the radius.
2. Use Pythagoras: $OT^2 + PT^2 = OP^2$.

Derivation:

$$\begin{aligned}OT^2 + 10^2 &= 26^2 \\ \Rightarrow OT^2 + 100 &= 676 \\ \Rightarrow OT^2 &= 576 \\ \Rightarrow OT &= 24\text{cm}.\end{aligned}$$

Circles — Tangents to a circle

- 16. 16.** A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.

Answer:

$$AB + CD = AD + BC$$

Solution:

1. Use the property that lengths of tangents drawn from an external point to a circle are equal.
2. Write equations for tangents from vertices A, B, C, and D.
3. Add the equations and group the terms to form the sides of the quadrilateral.

Derivation:

Let the circle touch sides AB, BC, CD, DA at P, Q, R, S respectively. $AP = AS, BP = BQ, CR = CQ,$

Circles — Tangents to a circle

- 17. 17.** From a point Q , the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. Find the radius of the circle.

Answer:

$$7\text{cm}$$

Solution:

1. Identify that the radius is perpendicular to the tangent at the point of contact.
2. This forms a right-angled triangle with the radius, tangent, and distance from the center as sides.
3. Apply the Pythagorean theorem to solve for the radius.

Derivation:

$$\begin{aligned}r^2 + 24^2 &= 25^2 \\ \Rightarrow r^2 + 576 &= 625 \\ \Rightarrow r^2 &= 49 \\ \Rightarrow r &= 7\text{ cm}.\end{aligned}$$

Circles — Tangents to a circle

- 18. 18.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8\text{cm}$$

Solution:

1. Draw a radius to the point of contact on the smaller circle, which acts as the perpendicular bisector of the chord.
2. Create a right triangle using the radius of the small circle, half the chord, and the radius of the large circle as the hypotenuse.
3. Calculate half the chord length and multiply by 2.

Derivation:

$$\text{Half-chord} = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} = 4 \text{ cm. Total length} = 2 \times 4 = 8 \text{ cm.}$$

Circles — Tangents to a circle

- 19. 19.** If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80° , then find $\angle POA$.

Answer:

$$50^\circ$$

Solution:

1. Recognize that the line from the center to the external point bisects the angle between the tangents.
2. Consider the quadrilateral $OAPB$ where angles at A and B are 90° .
3. Determine the angle $\angle APB$ is split in half and use the sum of angles in triangle OAP .

Derivation:

$$\begin{aligned} \angle APB &= 80^\circ \\ \Rightarrow \angle OPA &= 40^\circ. \text{ In } \triangle OAP, \angle OAP = 90^\circ. \angle POA = 180^\circ - 90^\circ - 40^\circ = 50^\circ. \end{aligned}$$

Circles — Tangents to a circle

- 20. 20.** Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line segment joining the points of contact at the centre.

Answer:

$$\text{Sum is } 180^\circ$$

Solution:

1. Let P be the external point and A, B be points of contact.
2. Consider the quadrilateral $OAPB$.
3. Use the property that the sum of angles in a quadrilateral is 360° and tangents are perpendicular to radii.

Derivation:

$$\begin{aligned} \angle OAP &= 90^\circ, \angle OBP = 90^\circ. \angle APB + \angle OAP + \angle OBP + \angle AOB = 360^\circ \\ \Rightarrow \angle APB + 90^\circ + 90^\circ + \angle AOB &= 360^\circ \\ \Rightarrow \angle APB + \angle AOB &= 180^\circ. \end{aligned}$$

Circles — Tangents to a circle